

# Lagged Causal Signals in Industrial Failure Diagnosis: An Exploratory Simulation Study

Independent Research Project

Anastasia Demidova

2026

## Abstract

This report examines how lag-aware statistical analysis can support root-cause diagnosis in industrial machine failures. I develop a synthetic multivariate time-series simulation of an industrial machine operating under changing regimes, with noisy sensor measurements, maintenance delay events, gradual component wear and rare failure events. The study compares a naive correlation-based failure ranking with lag-aware dependency scoring and Granger-style time-ordered analysis. The results show that contemporaneous correlations with failure are weak and difficult to interpret, while lag-aware signals recover several expected degradation-related relationships, including load  $\rightarrow$  temperature, temperature  $\rightarrow$  wear, wear  $\rightarrow$  vibration and wear  $\rightarrow$  failure. A logistic regression model with lagged features achieves a test ROC AUC of 0.804 under a time-based train-test split, confirming that temporal sensor patterns contain meaningful predictive information. The study demonstrates that even simple lag-aware methods can provide a more structured view of candidate root-cause signals than correlation alone, while also highlighting the limitations of exploratory causal interpretation from synthetic observational time series.

## 1. Introduction

Industrial machines generate large volumes of operational data, including sensor streams, operating modes, maintenance events and failure records. However, when a failure occurs, identifying the underlying reason often remains a manual and expert-driven process. A predictive model may indicate that a failure is likely, but it does not necessarily explain whether a variable is an upstream driver, an intermediate degradation mechanism or a late symptom.

This report develops a small simulation study of industrial failure diagnosis using synthetic multivariate time-series data. The purpose is not to build a complete causal discovery framework, but to examine whether lag-aware statistical signals can provide a more informative diagnostic view than simple contemporaneous correlation.

The study is motivated by three questions. First, how informative is a naive correlation-based ranking when failures are rare and binary? Second, can lag-aware statistical analysis recover meaningful time-ordered relationships in a simulated degradation process? Third, can candidate root-cause signals be distinguished from late-stage symptoms in a transparent and interpretable way?

The report makes three contributions. First, it develops a synthetic industrial time-series setting with operating regimes, degradation variables, maintenance delay and rare failures. Second, it compares correlation-based diagnosis with lag-aware dependency and Granger-style analysis. Third, it shows that simple time-aware methods can recover part of the underlying degradation structure, while remaining limited as causal discovery tools.

## 2. Simulation Design

### 2.1 Industrial system structure

The synthetic system represents a simplified industrial machine operating over 1,500 time steps. The machine switches between three operating regimes: normal operation, high-load operation and unstable operation. These regimes influence operational variables such as load, speed and pressure.

The system also includes degradation-related variables. Temperature responds to previous operating conditions. Wear accumulates gradually as a result of temperature, maintenance delay and load. Vibration acts as a late-stage symptom of accumulated wear. Failure events occur when the simulated risk becomes sufficiently high.

The simplified causal logic used in the simulation is:

```

operating_regime → load / speed / pressure
load, speed, pressure → temperature
temperature, maintenance_delay, load → wear
wear, pressure → vibration
pressure, temperature, maintenance_delay, wear, vibration → failure

```

The purpose of the simulation is not to reproduce a real machine in detail, but to create a controlled setting where the underlying dependency structure is known. This makes it possible to evaluate whether exploratory time-aware methods recover meaningful parts of the simulated mechanism.

The main degradation-related variables are temperature, maintenance delay, wear and vibration. Temperature reflects operational stress, maintenance delay contributes to accumulated degradation, wear represents the underlying degradation state, and vibration acts as a late observable symptom.

### 2.2 Dataset summary

The generated dataset contains 45 failure events, corresponding to a failure rate of 3%. This creates an imbalanced setting that is closer to real industrial failure diagnosis than a balanced classification problem.

*Table 1. Summary of the simulated dataset*

| Metric     | Value |
|------------|-------|
| Time steps | 1500  |

|                        |       |
|------------------------|-------|
| Failure events         | 45    |
| Failure rate           | 0.030 |
| Normal regime share    | 0.552 |
| High-load regime share | 0.321 |
| Unstable regime share  | 0.127 |

The simulated setting combines rare failure events with changing operating regimes, creating a simple but non-stationary diagnostic environment.

### **3. Methodology**

#### **3.1 Correlation-based baseline**

The first baseline computes absolute contemporaneous correlations between each sensor variable and the binary failure indicator. This approach is intentionally simple. It does not account for time delays, directionality or intermediate degradation mechanisms.

The baseline is used to illustrate the limitations of relying only on same-time associations in rare-event failure diagnosis.

#### **3.2 Lagged dependency scoring**

For each pair of candidate variables, lagged correlations are computed between  $\text{source}(t - \text{lag})$  and  $\text{target}(t)$  for lags from 1 to 5. The maximum absolute lagged correlation is used as an exploratory time-ordered dependency score.

This score does not prove causality. It is used as a transparent way to detect whether past values of one variable are informative about future values of another variable.

#### **3.3 Granger-style analysis**

A Granger-style test is also applied to candidate variable pairs. The test asks whether past values of a source variable help predict the current value of a target variable beyond the target's own history.

In this report, Granger-style results are interpreted as time-ordered predictive signals, not as definitive causal evidence. This distinction is important because observational time-series data may contain hidden confounding, feedback loops and non-stationarity.

#### **3.4 Root-cause signal ranking**

For candidate variables pointing to failure, lagged dependency information and Granger-style evidence are combined into an exploratory root-cause signal score. The purpose is to rank variables that may represent upstream signals, intermediate degradation factors or late-stage symptoms.

#### **3.5 Predictive baseline**

A logistic regression model with lagged features is trained as a predictive baseline. The model uses lagged values of sensor variables and operating regime indicators. To avoid evaluating the model on the same data used for training, a time-based 70/30 train-test split is used.

The predictive model is not interpreted causally. It is included to check whether lagged sensor information contains learnable signals about future failures.

## 4. Results

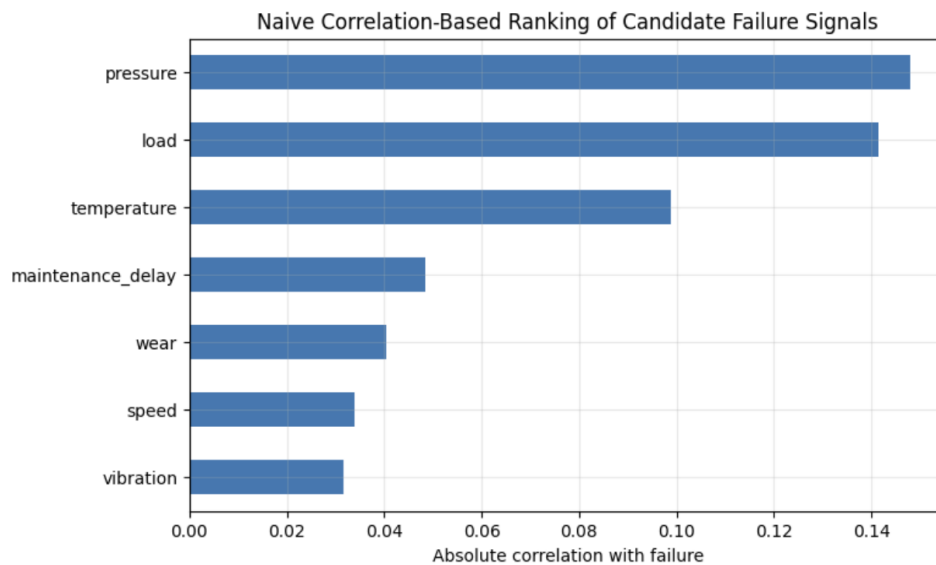
### 4.1 Main findings

The simulation results highlight three findings. First, naive contemporaneous correlation provides only weak diagnostic information. Second, lag-aware analysis recovers several meaningful time-ordered relationships from the simulated degradation process. Third, the root-cause signal ranking gives a more structured interpretation of failure mechanisms than correlation alone.

#### Finding 1. Contemporaneous correlations with failure are weak

The naive correlation baseline ranks pressure, load and temperature as the variables most associated with failure. However, all absolute correlations are relatively small. The highest correlation is only 0.148 for pressure, followed by 0.141 for load and 0.099 for temperature.

This result is expected. Failure is a rare binary event, and the simulated mechanism unfolds over time rather than through immediate same-time associations. A variable may be important for failure development without having a strong contemporaneous correlation with the failure indicator.



*Figure 1. Naive correlation-based ranking of candidate failure signals*

Figure 1 shows absolute contemporaneous correlations between sensor variables and failure events. The weak correlations suggest that root-cause diagnosis cannot rely only on same-time associations.

## **Finding 2. Lag-aware analysis recovers meaningful time-ordered signals**

Lag-aware dependency scoring recovers several expected relationships from the simulated degradation process. The strongest recovered signals include wear → vibration, load → temperature, pressure → temperature and temperature → wear.

These links are consistent with the design of the simulated system. The relationship wear → vibration reflects vibration as a late symptom of component degradation. The link temperature → wear reflects delayed degradation accumulation following elevated temperature. The link load → temperature captures the effect of operating intensity on thermal stress.

The Granger-style analysis supports a similar interpretation. It identifies strong time-ordered predictive signals such as load → temperature, temperature → wear, wear → vibration and wear → failure. These results are meaningful in the context of the simulation because they correspond to the intended degradation chain. However, they should be interpreted as time-ordered predictive evidence rather than proof of causality.

## **Finding 3. Root-cause signal ranking is more informative than correlation alone**

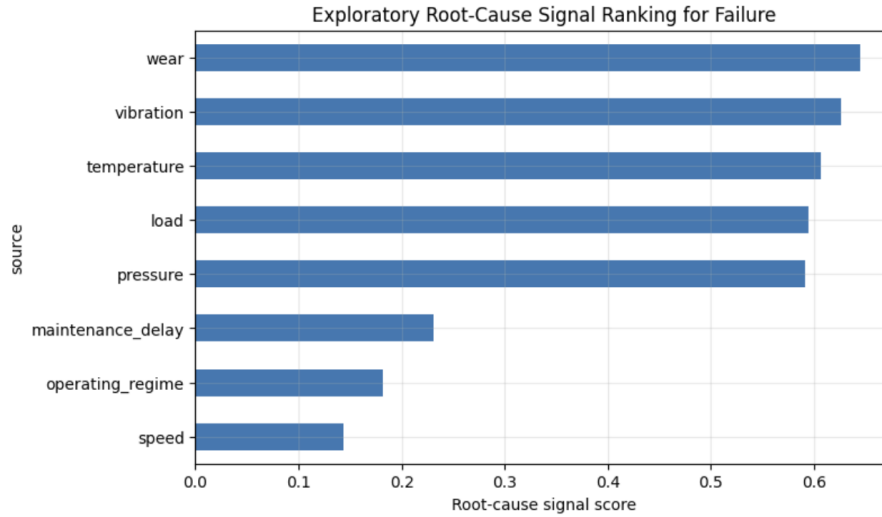
The combined exploratory ranking places degradation-related variables at the top.

*Table 2. Top candidate root-cause signals for failure*

| Source      | Best lag | Root-cause signal score |
|-------------|----------|-------------------------|
| wear        | 1        | 0.645                   |
| vibration   | 1        | 0.626                   |
| temperature | 2        | 0.607                   |
| load        | 4        | 0.595                   |
| pressure    | 3        | 0.591                   |

This ranking is more informative than the correlation baseline. It highlights wear as the strongest degradation-related signal and vibration as a late-stage symptom. Upstream operating variables such as temperature, load and pressure also receive high scores, reflecting their role in the simulated failure mechanism.

The comparison between the two rankings is important. The correlation baseline mainly captures weak same-time associations with the failure indicator. The lag-aware ranking, by contrast, better reflects the simulated degradation chain and provides a more structured diagnostic view.



*Figure 2. Exploratory root-cause signal ranking for failure*

Figure 2 ranks variables by their combined lag-aware and Granger-style signal score. Unlike simple correlation, the ranking highlights degradation-related variables such as wear and vibration, as well as upstream operating signals.

#### 4.2 Predictive validation

The logistic regression model with lagged features achieves a test ROC AUC of 0.804 under a time-based 70/30 train-test split. This indicates that lagged sensor features contain meaningful predictive information about future failures.

The most influential lagged features include vibration, wear, temperature, operating regime, maintenance delay and pressure. This supports the idea that temporal structure is important for failure diagnosis. However, prediction alone is not equivalent to explanation: a predictive model can identify useful features without distinguishing actionable upstream causes from late symptoms.

#### 4.3 Local failure case study

A local pre-failure explanation is generated for a failure event at time 389. The analysis compares the average value of each key degradation-related variable shortly before the failure with an earlier baseline window.

*Table 3. Local pre-failure changes before failure at time 389*

| Variable    | Baseline mean | Pre-failure mean | Change |
|-------------|---------------|------------------|--------|
| temperature | 0.666         | 3.603            | 2.938  |
| vibration   | 8.285         | 10.338           | 2.053  |
| wear        | 5.918         | 7.686            | 1.768  |

In this local case, temperature, vibration and wear increase substantially before the failure. This is consistent with a degradation-driven failure mechanism: elevated operating stress contributes to wear, and vibration increases as a late symptom before failure.



**Figure 3.** Local changes in degradation-related variables before the failure at time 389

Figure 3 shows the local evolution of temperature, maintenance delay, wear and vibration before and shortly after a failure event. Temperature, wear and vibration increase before the failure, supporting the interpretation of a degradation-driven failure pattern.

#### 4.4 Summary of exploratory recovery

The estimated top-10 time-ordered links recover 6 links from the simplified true structure. This corresponds to a precision of 0.600 and a recall of 0.375. These values are appropriate for an exploratory method: the results show that simple lag-aware analysis can recover part of the simulated dependency structure, but it is not sufficient for reliable causal discovery in complex real-world systems.

### 5. Discussion

The results show that industrial failure diagnosis should not rely only on contemporaneous correlation. In this simulation, direct correlations with failure are weak, partly because failure is a rare binary event and partly because the failure mechanism unfolds through delayed dependencies.

The lag-aware analysis provides a more structured diagnostic view. It recovers several expected time-ordered relationships, including load → temperature, temperature → wear, wear → vibration and wear → failure. These links align with the simulated degradation chain and help distinguish upstream operating signals, intermediate degradation variables and late-stage symptoms.

The comparison between correlation-based and lag-aware rankings is especially important. The correlation baseline ranks pressure and load most highly, but it gives low scores to wear and vibration. In contrast, the lag-aware root-cause signal ranking places wear and vibration at the top, while still recognising the relevance of temperature, load and pressure. This suggests that temporal structure can provide a more informative basis for root-cause analysis than same-time association alone.

The predictive baseline further supports this conclusion. A logistic regression model using lagged features achieves a test ROC AUC of 0.804, indicating that failure events are partially predictable from historical sensor behaviour. However, prediction alone is not equivalent to explanation. The model can identify useful features, but additional causal reasoning is required to distinguish actionable causes from symptoms.

The study also has clear limitations. The dataset is synthetic and simplified. The causal structure is known by design, whereas real industrial systems often contain hidden variables, incomplete logs, changing operating conditions and measurement errors. Granger-style tests indicate time-ordered predictive signals, not definitive causality. In addition, the current approach does not estimate causal effect sizes or integrate engineering knowledge.

These limitations point to several natural extensions. Future work could test more rigorous causal discovery methods, incorporate domain constraints from engineering knowledge, model non-stationary operating conditions, include text-based maintenance logs or estimate causal effects using methods such as Double Machine Learning or Causal Forests. The present study should therefore be interpreted as an exploratory computational step rather than a complete industrial diagnostic framework.

## **6. Conclusion**

This report developed a small simulation study of lagged causal signals in industrial failure diagnosis. The synthetic system included changing operating regimes, noisy sensor data, gradual component wear, maintenance delay and rare failure events.

The main finding is that simple correlation-based diagnosis is insufficient for identifying meaningful root-cause signals in time-dependent failure processes. Lag-aware analysis provides a more structured view by recovering expected relationships in the simulated degradation chain and by ranking wear, vibration, temperature, load and pressure as important candidate signals.

The predictive baseline confirms that lagged sensor features contain useful information about future failures, achieving a test ROC AUC of 0.804. At the same time, moderate graph recovery precision and recall show that simple exploratory methods cannot replace rigorous causal discovery.

Overall, the project demonstrates how synthetic simulation, time-series analysis and cautious causal interpretation can be combined into a transparent computational study of industrial root-cause diagnosis.